

作业 5

习四 18, 21, 22, 27, 29

18.

$$(1) \quad N(0) \text{ 强度 } \int_0^1 u du = \frac{1}{2} \times 4 = 2.$$

$$P(N(1)=3) = \frac{2^3}{3!} e^{-2} = \frac{8}{6} e^{-2} = \frac{4}{3} e^{-2}$$

19

$$N(2) - N(1) \text{ 强度 } \int_1^2 u du = \frac{1}{2}.$$

$$N(2) - N(1) \text{ 强度 } \int_1^2 u du = \frac{4-1}{2} = \frac{3}{2}.$$

$$P(N(1)=2, N(2)=4)$$

$$= P(N(1)=2) P(N(2)-N(1)=2)$$

$$= \frac{(2)^2}{2!} e^{-2} \cdot \frac{(3)^2}{2!} e^{-\frac{3}{2}}$$

$$= \frac{1}{8} \times \frac{9}{8} e^{-2}$$

$$= \frac{9}{64} e^{-2}$$

(3)

$$P = C_4^2 \left(\frac{1}{4}\right)^2 \left(1 - \frac{1}{4}\right)^2$$

$$= 6 \times \frac{1}{16} \times \frac{9}{16}$$

$$= \frac{27}{128}$$

21.

$$(1) \quad P\left\{B(3.6) - B(2.39) \leq 0.9 \mid B(1.6) = 0.8, B(2.39) = -0.1\right\}$$

$$= P(B(3.6) - B(2.39) \leq 0.9)$$

$$= \Phi\left(\frac{0.9}{\sqrt{1.21}}\right) = \Phi\left(\frac{0.9}{1.1}\right)$$

$$= \Phi(0.81)$$

$$(2) \quad \text{Cov}(B(8) - B(4), B(4))$$

$$= \text{Cov}(B(8) - B(4) + B(4) - B(4), B(4))$$

$$= \text{Cov}(B(8), B(4)) - \text{Cov}(B(4), B(4))$$

$$= 4 - 4$$

$$= 0$$

$$(3) \quad D(B(1) + B(2))$$

$$= 4 D(B(1)) + 2 \text{Cov}(B(1), B(2)) + D(B(2))$$

$$= 4 + 2 + 4 \times 1$$

$$= 10$$

22.

$$\textcircled{1} \quad X(0) = 0, \quad X(t) = (t+1)B\left(\frac{1}{t+1}\right) - B(1) = B(t+1) - B(1)$$

$$X(t) = 1 - 1 = 0$$

$$D_X(t) =$$

$$X(t)$$

$$\textcircled{2} \quad 0 \leq s \leq t$$

$$X(t) - X(s) = (t+1)B\left(\frac{1}{t+1}\right) - (s+1)B\left(\frac{1}{s+1}\right)$$

$$= B(t+1) - B(s+1)$$

$$= B(t-s) \sim N(0, t-s).$$

增量平稳正态.

$$\textcircled{3} \quad \text{增量独立} \quad 0 \leq t_1 \leq t_2 \leq \dots \leq t_n$$

$$\text{if } k, \quad \text{Cov}(X_{i+1} - X_i, X_{k+1} - X_k)$$

$$= \text{Cov}$$

22.

$$X(t) = (t+1)B\left(\frac{1}{t+1}\right) - B(1)$$

$$= B(t+1) - B(1)$$

$$= B(t) \text{ 标准布朗运动}$$

27.

(1)

$$B(\frac{1}{4}) = 2 \Leftrightarrow \frac{1}{4} B(1) = 2 \quad B(1) = 8$$

$$B(\frac{1}{4}) = \frac{12}{5} \Leftrightarrow B(1) = \frac{48}{5}$$

$$B(\frac{1}{4}) > \frac{3}{2} \Leftrightarrow B(1) > 6$$

PT

$$P_1 = P(B(1) - B(0) \geq 3)$$

$$= 1 - \Phi\left(\frac{3}{\sqrt{2}}\right)$$

$$= 0.0668$$

(2)

$$P(B(1) \leq x \mid B(0) = 2, B(\frac{1}{4}) = \frac{12}{5})$$

$$= P(B(1) \leq x \mid B(0) = 12, B(\frac{1}{4}) = \frac{48}{5})$$

$$= P(B(1) - B(0) \leq x - 12)$$

$$= \Phi\left(\frac{x - 12}{\sqrt{2}}\right)$$

$$F(x) = \Phi\left(\frac{x - 12}{\sqrt{2}}\right)$$

$$\mu = 12$$

$$\sigma^2 = \left(\frac{1}{\sqrt{2}}\right)^2 = 0.04$$

$$B(1) \sim N(12, 0.04)$$

29.

$$(1) P_1 = P(-x \leq B(t) \leq x)$$

$$= \Phi\left(\frac{x}{\sqrt{t}}\right) - \Phi\left(\frac{-x}{\sqrt{t}}\right)$$

$$= 2\Phi\left(\frac{x}{\sqrt{t}}\right) - 1$$

(2)

$$P(\max_{0 \leq t \leq T} B(t) \leq x)$$

$$= P(\max_{0 \leq t \leq T} B(t) \leq x)$$

$$= 1 - 2P(B(t) > x)$$

$$= 1 - 2[1 - \Phi\left(\frac{x}{\sqrt{t}}\right)]$$

$$= 2\Phi\left(\frac{x}{\sqrt{t}}\right) - 1$$